

MARIA ROSABELLE (“ROSS”) ONG, Ph.D.

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Marine & Aquatic Biogeochemistry | Climate and Ocean Sciences | Environmental Technology

EDUCATION

Ph. D. Earth and Environmental Sciences, *Columbia University*, New York, NY, USA, Oct. 2025

M. Phil. Earth and Environmental Sciences, *Columbia University*, New York, NY, USA, Feb. 2023

M.A. Earth and Environmental Sciences, *Columbia University*, New York, NY, USA, Jun. 2021

B.S. Biology, *cum laude*, *University of the Philippines Manila*, Manila, Philippines, Oct. 2007

RESEARCH EXPERIENCE

July 2026 – present **Founder**, *Aetherium Advisory*, Singapore, Singapore

Aug. 2026 – present **Research Associate**, *American Museum of Natural History*, New York, NY, USA

Feb. 2026 – Aug. 2026 **Postdoctoral Scientist**, *American Museum of Natural History*, New York, NY, USA

Sep. 2019 – Oct. 2025 **Ph.D. Candidate**, *Columbia University*, *Lamont Doherty Earth Observatory*, *American Museum of Natural History*, New York, NY, USA

Dec. 2015 – Aug. 2019 **Research Assistant**, *Asian School of the Environment / Earth Observatory of Singapore*, *Nanyang Technological University*, Singapore

PROFESSIONAL EXPERIENCE

Oct. 2014 – Dec. 2015 **Water Quality Analyst**, *Resorts World Sentosa Pte. Ltd.*, Singapore

Nov. 2011 – Apr. 2014 **Service Manager / Executive**, *CPG FM Pte. Ltd.*, Singapore

PUBLICATIONS (IN REFEREED JOURNALS)

Mahowald, N., Liddy, H., [and 35 others including Ong, M. R.] (2026). Characterizing Natural Aerosol Variability and its Uncertainty in the Earth System. *Atmospheric Chemistry and Physics*. (In Review).

Jung, J., Duprey, N. N., [and 40 others including Ong, M. R.] (2025). Equatorial upwelling of phosphorus Drives Atlantic N₂ fixation and *Sargassum* blooms. *Nature Geoscience*.

<https://doi.org/10.1038/s41561-025-01812-2>

- Highlighted in an editorial article by Nature Geoscience (2025), <https://doi.org/10.1038/s41561-025-01886-y>; and in a News and Views article in Nature Geoscience, Baker and Wang (2025), <https://doi.org/10.1038/s41561-025-01871-5>

Walter, R. M., Sayani, H. R., [and 28 others, including Ong, M. R.] (2023). The CoralHydro2K Database: a global, actively curated compilation of coral $\delta^{18}\text{O}$ and Sr/Ca proxy records of tropical ocean hydrology and temperature for the Common Era. *Earth System Science Data*, 15(5), 2081-2116. <https://doi.org/10.5194/essd-15-2081-2023>

Ong, M. R., Goodkin, N. F., Guppy, R., Huguen, K. A. (2022). *Colpophyllia natans* from Tobago, a novel paleoclimate archive for reconstructing sea surface temperatures in the tropical Atlantic. *Paleoceanography and Paleoclimatology*, 37(12), e2022PA004483. <https://doi.org/10.1029/2022PA004483>

- Featured and highlighted in a commentary by Thirumalai and Maupin (2023), <https://doi.org/10.1029/2023PA004723>

Kaushal, N., Tanzil, J. T. I., Zhou, Y., **Ong, M. R.**, Goodkin, N. F., Martin, P. (2022). Environmental Calibration of Coral Luminescence as a Proxy for Terrigenous Dissolved Organic Carbon Concentration in Tropical Coastal Oceans. *Geochemistry, Geophysics, Geosystems*, 23(10), e2022GC010529. <https://doi.org/10.1029/2022GC010529>

Goodkin, N. F., Samanta, D., Bolton, A., **Ong, M. R.**, Phan, K. H., Vo, S. T., Karnauskas, K. B., Hughen, K. A., (2021). Natural and Anthropogenic Forcing of Multi-decadal to Centennial Scale Variability of Sea Surface Temperature in the South China Sea. *Paleoceanography and Paleoclimatology*, 36(10), e2021PA004233. <https://doi.org/10.1029/2021PA004233>

Goodkin, N. F., Bolton, A., Hughen, K. A., Karnauskas, K. B., Griffin, S., Phan, K. H., Vo, S. T., **Ong, M. R.**, Druffel, E. R. M. (2019). East Asian Monsoon variability since the sixteenth century. *Geophysical Research Letters*, 46(9), 4790-4798. <https://doi.org/10.1029/2019GL081939>

He, S., Goodkin, N. F., Jackisch, D., **Ong, M. R.**, Samanta, D. (2018). Continuous real-time analysis of the isotopic composition of precipitation during tropical rain events: Insights into tropical convection. *Hydrological Processes*, 32(11), 1531-1545. <https://doi.org/10.1002/hyp.11520>

PUBLICATIONS (In-Preparation)

Ong, M. R., Goodkin, N. F., Guppy, R., Hughen, K. A. (*in preparation*). Western Tropical Atlantic Hydroclimate Reveal Interannual ENSO-AMM Drought Patterns and Interhemispheric SST Gradient-Driven ITCZ Shifts

Ong, M. R., Goodkin, N. F., Guppy, R., Hu, H. M., Shen, C. C., Hughen, K. A. (*in preparation*). Non-Stationary Interdecadal Variability in the North Atlantic Oscillation Revealed in a 185-Year Coral-Based Sea Surface Temperature Record from the Western Tropical North Atlantic

Ong, M. R., Goodkin, N. F., Gordon, A. L., Guppy, R., Hughen, K. A. (*in preparation*). Coral $\Delta^{14}\text{C}$ from Tobago Reveals a Possible Shift in Surface Water Circulation in the Tropical North Atlantic since the early 2000s.

DATASET CONTRIBUTIONS

Jung, J., Duprey, N. N., [and 40 others including **Ong, M. R.**] (2025). Equatorial upwelling of phosphorus drives Atlantic N₂ fixation and Sargassum blooms [Dataset]. Dryad. <https://doi.org/10.5061/dryad.jm63xsjkq>

Ong, M. R., Goodkin, N. F., Guppy, R., Hughen, K. A. (2022). NOAA/WDS Paleoclimatology – Tobago coral Sr/Ca, extension and SST data from 1988-2016 CE [Dataset]. NOAA National Center for Environmental Information. <https://doi.org/10.25921/9xbw-sk21>

HONORS AND AWARDS

2026	WHOI Postdoctoral Scholarship , Woods Hole Oceanographic Institution (\$131,000) (<i>offered, declined</i>)
2023	Sydney Anderson Travel Award , Richard Gilder Graduate School, American Museum of Natural History. (\$2,500)
2020 – 2021	Kenneth and Linda Ciriacks Graduate Fellowship in Earth and Environmental Sciences , Columbia University (GSAS).

2019	Lerner-Gray Fund for Marine Research , <i>American Museum of Natural History</i> .
2019	Richard Gilder Graduate School Graduate Fellow , <i>American Museum of Natural History</i> .
2019	Dean's Fellow , <i>Department of Earth and Environmental Sciences, Columbia University</i> .

RESEARCH GRANTS AWARDED

01/2023-12/2024	Goodkin, N. F. (Lead PI) and Ong, M. R. (Co-PI). Reconstructing Florida Current transport to better understand Atlantic Meridional Overturning Circulation (AMOC) behavior in a warming world. <i>LDEO – The Climate Center</i> . \$12,000.
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CONFERENCE PRESENTATIONS / ABSTRACTS (*Oral Presentation)

Humphries, C., Goodkin, N. F., **Ong, M. R.**, Ramos, R., and Siringan, F. P. Coral Ba/Ca as a Proxy for Wet Season Precipitation at Palau Island, Philippines. San Francisco, USA. Dec. 7-11, 2026. (*upcoming*)

Goodkin, N. F., **Ong, M. R.**, Tay, J., and Kannad, A. Marine Carbonate Archives Capture a Non-Stationary North Atlantic Oscillation Footprint. *AGU Fall Meeting 2026*. San Francisco, USA. Dec. 7-11, 2026. (*upcoming*)

Ong, M. R., Goodkin, N. F., Guppy, R., Hu, H. M., Shen, C. C., and Hughen, K. A. Non-Stationary Interdecadal Variability in the North Atlantic Oscillation Revealed in a 185-Year Coral-Based Sea Surface Temperature Record from the Western Tropical North Atlantic. *EGU General Assembly 2026*. Vienna, Austria. May 3-8, 2026.

Ong, M. R., Goodkin, N. F., Guppy, R., Hughen, K. A. Annual Coral $\Delta^{14}\text{C}$ from Tobago Reveals a Possible Shift in Surface Water Circulation in the Tropical North Atlantic since the early 2000s. *AGU Fall Meeting 2024*. Washington D.C., USA. Dec. 9-13, 2024

Ong, M. R., Goodkin, N. F., Guppy, R., Hughen, K. A. A Coral-based Multiproxy Investigation of Regional Hydroclimate Variability in the Tropical Atlantic since the 1960s. *Ocean Sciences Meeting 2024*. New Orleans, LA, USA. Feb. 18-23, 2024.

Humphries, C., Goodkin, N. F., Brenner, L. D., **Ong, M. R.**, Ramos, R. D. Potential use of coral Ba/Ca off Palau Island, Philippines, as a proxy to study past climate and investigate the impacts of ENSO. *Ocean Sciences Meeting 2024*. New Orleans, LA, USA. Feb. 18-23, 2024.

Goodkin, N. F.*, Ramos, R. D., Murty, S. A., Samanta, D., Chen, M., Kannad, A., **Ong, M. R.**, and the MarGe Laboratory Group. An Integrated Review of the Climate Driven Variations in the Indo-Pacific using a Multi-Proxy, Multi-Geographical Approach in Scleractinian Corals. *AGU Fall Meeting 2023*. San Francisco, CA, USA. Dec. 11-15, 2023.

Ong, M. R., Goodkin, N. F., Fraser, A. G., Guppy, R., Hughen, K. A. Investigating the Utility of Coral Ba/Ca from Tobago as a Proxy for River Discharge and Regional Hydroclimate Variability in the Tropical Atlantic. *AGU Fall Meeting 2022*. Chicago, IL, USA. Dec. 12-16, 2022.

Goodkin, N. F., Kannad, A., Yambing, J. R., **Ong, M. R.**, Kilbourne, K. H., Hughen, K. A. Examination of the North Atlantic Oscillation and the Ability to Improve Marine Reconstructions Using Spatially Diverse Coral Records. *AGU Fall Meeting 2022*. Chicago, IL, USA. Dec. 12-16, 2022.

***Ong, M. R.**, Goodkin, N. F., Guppy, R., Hughen, K. A. *Colpophyllia natans*, a potential new paleoclimate

- archive for reconstructing sea surface temperatures in the tropical North Atlantic. *Ocean Sciences Meeting 2022* (Virtual). Feb. 24 – Mar. 4, 2022.
- Goodkin, N. F., Samanta, D., Bolton, A., **Ong, M. R.**, Hoang, P. K., Vo, S.T., Karnauskas, K. B., Hughen, K. A. A novel identification of anthropogenic changes in South China Sea seasonal sea surface temperatures over the past 400 years. *Ocean Sciences Meeting 2022* (Virtual). Feb. 24 – Mar. 4, 2022.
- ***Ong, M. R.**, Tan, A. S. Y., Gan, M. C., Kho, P., Tanzil, J. T. I., Lee, J. N., Goodkin, N. F. Annual coral $\delta^{18}\text{O}$ variability from the Singapore-Malay Peninsula reveals impacts from ENSO. *14th International Coral Reef Symposium (ICRS)*. Bremen, Germany (Virtual). Jul. 18-23, 2021 (*Abstract accepted and withdrawn*)
- *Goodkin, N. F., Samanta, D., Hughen, K. A., **Ong, M. R.**, Bolton, A., Karnauskas, K. Vietnam climate experienced multi-decadal variability over the past 400 years and decoupling of seasonal SST with anthropogenic warming. *14th International Coral Reef Symposium (ICRS)*. Bremen, Germany (Virtual). Jul. 18-23, 2021
- ***Ong, M. R.**, Reconstructing Long-Term Climate Variability using Slow Growing Brain Corals from the Caribbean. *12th Annual RGGS Second Year Student Symposium*. Virtual Presentation. Dec. 10, 2020.
- ***Ong, M. R.**, Reconstructing Long-Term Climate Variability using Slow Growing Brain Corals from the Caribbean. *LDEO First Year Colloquium*. Virtual Presentation. Apr. 17, 2020.
- Ong, M. R.**, Tanzil, J. T. I., Goodkin, N. F. Coral $\delta^{18}\text{O}$ reconstructs sea surface salinity in the Singapore Straits revealing impacts from ENSO. *MSRDP Mid-term Symposium*. Nanyang University of Singapore, Singapore. Feb. 21-22, 2019.
- Ong, M. R.**, Tanzil, J. T. I., Goodkin, N. F. Coral $\delta^{18}\text{O}$ reconstructs sea surface salinity in the Singapore Straits revealing impacts from ENSO. *EOS 2019 Scientific Annual Meeting*. Nanyang Technological University, Singapore. Jan. 10-11, 2019
- Ong, M. R.**, Tanzil, J. T. I., Goodkin, N. F. Coral $\delta^{18}\text{O}$ reconstructs sea surface salinity in the Singapore Straits revealing impacts from ENSO. *AGU Fall Meeting 2018*. Washington D.C., USA. Dec. 10-14, 2018.
- Loke, D, Goodkin, N. F., Bolton, A., **Ong, M. R.** 240-Year record of Sr/Ca and Ba/Ca from a coral off central Vietnam reveals impacts of monsoonal upwelling and La Niña rains. *AGU Fall Meeting 2018*. Washington D.C., USA. Dec. 10-14, 2018.
- ***Ong, M. R.**, Tanzil, J. T. I., Goodkin, N. F. Using coral Sr/Ca and $\delta^{18}\text{O}$ to track environmental changes in the Singapore Straits. *4th Asia Pacific Coral Reef Symposium*. Cebu, Philippines. Jun. 04-08, 2018.
- He, S., Goodkin, N.F., Jackisch, D., **Ong, M. R.** Continuous real-time measurements of δ -values of precipitation during rain events: Insights into tropical convection. *AGU Fall Meeting 2017*. New Orleans, LA, USA. Dec. 11-15, 2017.
- Jackisch, D., He, S., **Ong, M. R.**, Goodkin, N. F. Continuous and simultaneous measurements of precipitation and vapor isotopes over two monsoon seasons during 2016-2017 in Singapore. *AGU Fall Meeting 2017*. New Orleans, LA, USA. Dec. 11-15, 2017.
- Jackisch, D., He, S., **Ong, M. R.**, Goodkin, N. F. Simultaneous measurements of stable isotope composition in precipitation and vapour during NE monsoon in Singapore. *AOGS 14th Annual Meeting*. Singapore. Aug. 06-11, 2017.

He, S., Goodkin, N. F., Jackisch, D., **Ong, M. R.** Continuous real-time analysis of isotopic composition of precipitation during tropical rain events using a diffusion sampler. *Advances in Stable Isotope Techniques and Applications Conference*. Canada. June 2017.

INVITED PRESENTATIONS AND LECTURES (*Oral Presentation)

- ***Ong, M. R.**, Ocean Time Capsules: Corals as Recorders of Past Environments. *American Museum of Natural History, Physical Sciences Jamboree*. Keynote Speaker. Nov. 14, 2025.
- ***Ong, M. R.**, Untapping a New Climate Archive: *Colpophyllia natans* Climate Reconstructions in the Tropical Atlantic. *University of New Mexico – Center for Stable Isotopes Brown Bag Seminar*. Virtual Presentation. April 03, 2023.
- ***Ong, M. R.**, Decolonizing Paleontology: A Case Study. *Seminar in Race, Climate Change, Environmental Justice: On Geology's Colonial Legacies (Spring 2023)*. Guest Lecturer/Speaker. Feb. 16, 2023.
- ***Ong, M. R.**, Corals and Climate Change. *Lerner Gray Committee Meeting*. Virtual Presentation. May 05, 2020.
- ***Ong, M. R.**, Ramos, R. D., Corals as archives of past climate. *National Marine Biology and Museum and Aquarium, Checheng, Pingtung, Taiwan, R.O.C.*, Sep. 2017.

ACADEMIC SERVICE, SCIENTIFIC AND EDUCATIONAL OUTREACH

2022 – present	Journal Reviewer. AGU <i>Geochemistry, Geophysics, Geosystems</i> , AGU <i>Paleoceanography and Paleoclimatology</i> , Association for the Sciences of Limnology and Oceanography (ASLO), <i>Limnology and Oceanography: Methods</i> , EGU <i>Climate of the Past</i> , <i>Proceedings of the National Academy of Sciences (PNAS)</i>
May – Oct. 2026	Reviewer. IPCC AR7 Working Group 1 First Order Draft ECR Group Review – Chapter 6. Intergovernmental Panel on Climate Change.
Jan. 15, 2026	Speaker / Host. South Brooklyn High School Residency Program, American Museum of Natural History, New York, NY.
Dec. 17, 2025	Presenter / Co-Host (Paleoclimate Lab Tour). Academy For Teachers Masterclass on Corals, Oceans and Climate: Past, Present and Future by Nathalie Goodkin, American Museum of Natural History, New York, NY.
Nov. 2025	Reviewer. First-Time Presenter Feedback Program, American Geophysical Union.
Jul. 14, 2025	Panelist (Speaker). Lang Science Program, Designing and Presenting Scientific Posters (10 th Grade), American Museum of Natural History.
Apr. 11, 2025	Guest Scientist (Speaker). The Science and Nature Program, Fellowship of the Young Scientist (3 rd Grade), American Museum of Natural History.
Mar. 13, 2025	Speaker / Host. South Brooklyn High School Residency Program, American Museum of Natural History, New York, NY.
Fall 2024	Mentor. DEES Application Mentorship Program. Columbia University
Jun. 2023	Guest Reviewer / Judge. Billion Oyster Project's 9 th Annual Student Symposium, Billion Oyster Project. New York Harbor School, Governor's Island, New York, NY.
Apr. 2021	Chat Moderator. Frontiers Lecture: Unlocking Climate Data in Corals. AMNH EarthFest Online. Livestream on Zoom.
Apr. 2021	Chat Moderator. Astronomy Online: From Sun to Sea Life. AMNH EarthFest Online. Livestream on Zoom.

Aug. 2020	Speaker. <i>Corals and Climate Change</i> . American Museum of Natural History (AMNH) Learn with Me Online Program. Livestream on YouTube.
Summer 2020	Volunteer. <i>Lamont Summer Mentorship Program</i> , Columbia University.
Apr. 2020	Participant. <i>Rotunda Magazine</i> . Featured in an article (for field work done in Trinidad and Tobago funded by the Niarchos Foundation).

TEACHING EXPERIENCE

Oct. 13, 2025	Guest Lecturer (3-hour virtual Zoom lecture). Introduction to Paleooceanography, <i>Florida Atlantic University</i>
Spring 2022	Teaching Assistant. Chemistry of Continental Waters, <i>Columbia University</i>
Spring 2021	Teaching Assistant. Earth's Oceans and Atmosphere, <i>Columbia University</i>

FIELD EXPERIENCE

Jun. 2026	Participant. Coral reef survey. Fort Lauderdale, Florida, <i>USA</i> .
Jul. 2023	Participant. Coral core collection. Fort Lauderdale, Florida, <i>USA</i> .
Oct. 2019	Participant. Coral core collection. Kelleston Drain, Speyside and St. Giles Island, <i>Trinidad and Tobago</i> .
2016 – 2019	Participant. Coral core collection and water sampling. Southern Islands, <i>Singapore Straits</i> .

PROFESSIONAL DEVELOPMENT / WORKSHOPS / CONFERENCES BY APPLICATION

Jul. 6-10, 2026	<i>2026 CESM Tutorial Workshop</i> . NCAR, Boulder, CO
Apr. 8, 2026	<i>Machine Learning for Earth Systems Modelling – Course 1 Foundations and New Frontiers</i> . European Centre for Medium-Range Weather Forecasts (ECWMF). (Virtual).
Jan. 12-13, 2026	<i>LEAP Momentum Bootcamp in Climate Data Science</i> . Columbia University, New York, NY (Virtual).
Oct. 20-22, 2025	<i>AIMES/PAGES/IGAC Workshop on Historical and Paleoclimate Aerosol Radiative Effects</i> . Lamont-Doherty Earth Observatory, New York, USA
Jan. 29-31, 2025	<i>Marine Data for Ocean Health 2025</i> . Copernicus Marine Team (Virtual).
Dec. 8, 2024	<i>Python for Satellite Remote Sensing: Analysis and Visualization for Earth Scientist</i> . AGU Fall Meeting 2024, Washington D.C.
Dec. 8, 2024	<i>Introduction to Radiocarbon: Theory and Best Practices for Ocean and Earth Sciences</i> . AGU Fall Meeting 2024, Washington D.C.
Aug. 21-23, 2023	<i>Paleo Data Assimilation Workshop</i> . Northern Arizona University (Virtual).
Jun. 5-8, 2023	<i>Teaching Development Program Sprint – Foundational Track</i> . Center for Teaching and Learning (CTL), Columbia University.
Dec. 11, 2022	<i>Introducing the Community Earth System Model (CESM2) via an Interactive Hands-on Tutorial</i> . AGU Fall Meeting 2022, Chicago, IL
Jun. 20–Jul. 01, 2022	<i>ISOCAMP 2022</i> . University of New Mexico – Center for Stable Isotopes.

PROFESSIONAL AFFILIATIONS / MEMBERSHIPS

2026 – Present	Member , European Geophysical Union
2018 – 2025	Member , American Geophysical Union
2022 – Present	Member , The Oceanography Society
2020 – 2025	Member , International Coral Reef Society
Mar. 2018 – Present	Rescue Diver , Professional Association of Diving Instructors (PADI).
Sep. 2019 – Present	Diving First Aid for the Professional Diver Provider , DAN.

LABORATORY SKILLS

Experienced in chemical analysis using: (1) ThermoFisher iCAP 6000/7400 Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES), (2) Kiel IV Carbonate Device, (3) Thermo MAT 253/+ mass spectrometer, (4) Thermo Finnigan Gas Bench II, (5) Delta V Isotope Ratio Mass Spectrometer, and (6) Picarro Diffusion Sampler – Cavity Ring-Down Spectrometer.

COMPUTER SKILLS

Knowledge in using Python, MATLAB, Kaleidagraph, QGIS 3, and OceanDataView (ODV). Proficient in Microsoft Office Suite, Adobe Photoshop, QAnalyseries, PyAnalyseries, Datagraph. Well-versed in both MAC OS and Windows OS platforms.

STUDENTS SUPERVISED

Julian Tay	Student Intern / Volunteer, Phillips Academy Andover, AMNH, New York, NY, USA, June 2026
Kyle Chang	BSc, Science Volunteer, AMNH, New York, NY, USA, Fall 2024 – Spring 2025
Cate Humphries	BSc, Barnard College, New York, NY, USA, Summer 2023 – Spring 2024
Rosy Arora	BSc, Georgetown University, Washington D.C. USA, Summer 2022, Summer Intern
Alexandra G. Fraser	BSc, Bard College, New York, USA, Summer 2022, NSF REU Summer Internship Program at AMNH
Adam Vasquez	BSc, Vassar College, New York, USA, Summer 2022, NSF REU Summer Internship Program at AMNH
Ong Chai Kee	MSc, Universiti Malaysia Terengganu, Malaysia, Summer AY 2017-18
Syamil Sahar	MSc, Universiti Malaysia Sarawak, Malaysia, 2 nd Sem AY 2017-18, AY 2018-19
Tiffany Lee Wan Qi	BSc, Nanyang Technological University, Singapore, 2 nd Sem, AY 2017-18
Deborah Loke	BSc, Nanyang Technological University, Singapore, Final Year Undergraduate Student, AY 2017-18)
Wanling Goh	Student Intern, Singapore American School, Singapore Summer Internship, AY 2016-17
Revadi Devi Thyartan	MSc, Swinburne University of Technology, Malaysia, AY 2016-17
Aloysius Teng	BSc, National University of Singapore, Singapore, Undergraduate Summer & Fall Internship, AY 2015-16
Gan Min Chong	MSc, Universiti Malaysia Terengganu, Malaysia, 2 nd Sem AY 2015-16